

Supplementary Material

Three Mechanisms Inflate Reported Performance on a Widely Used Public Chronic Kidney Disease Benchmark

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S1. SCOPE OF THIS SUPPLEMENT

This document holds material that supports the main text but would displace its argument: the full feature registry, the complete provenance sensitivity suite, candidate flagging values for the audit framework, the extended discussion of synthetic-data pipelines, reproducibility detail, and the reporting checklist.

Every quantity is generated from a committed result file. Where an analysis could not be completed, this supplement says so rather than omitting the question.

S2. FEATURE-AVAILABILITY AND LEAKAGE REGISTRY

The registry (`data/feature_registry.csv`) records 18 fields for each of the 29 columns. Table S1 summarises the classification; the full file, including per-variable evidence and uncertainty notes, ships with the code.

TABLE S1
VARIABLES PER LEAKAGE CATEGORY.

Leakage category	Variables
Direct outcome copy	2
Deterministic derivative	2
KDIGO criterion input	2
Urine sediment (possible criterion)	4
Correlated renal biomarker	4
Consequence of advanced disease	4
Uncertain provenance	6
Legitimate pre-index	5

Feature configurations are derived from this file rather than maintained separately. `tests/test_registry.py` asserts that the derivation reproduces the configurations used in the reference analysis exactly; if a classification is edited without updating the configurations, or vice versa, the test suite fails.

S3. PROVENANCE SENSITIVITY: FULL RESULTS

Section V-D of the main text reports the headline sensitivity analyses. Table S2 gives the complete leave-one-variable-out results, generated from the result file rather than transcribed.

TABLE S2

LEAVE-ONE-VARIABLE-OUT MATCHING, COMPLETE RESULTS. EACH ROW REMOVES ONE CONTAINMENT VARIABLE AND RE-RUNS THE MATCH. THE MATCH FRACTION IS 1.000 IN EVERY CASE; *unique pins* IS THE DISCRIMINATING STATISTIC. ORDERED BY UNIQUE PINS, FEWEST FIRST.

Variable removed	k	Match	Unique pins
age	12	1.000	133
bgr	12	1.000	159
bu	12	1.000	170
al	12	1.000	171
sod	12	1.000	172
sg	12	1.000	179
hemo	12	1.000	179
su	12	1.000	180
sc	12	1.000	181
pcv	12	1.000	184
rbcc	12	1.000	184
wbcc	12	1.000	185
pot	12	1.000	187

A. Matched-pair audit

`table_45_matched_pair_audit.csv` contains twelve reproducibly sampled matched pairs (seed fixed in the script), each showing the analysed patient's published interval alongside the candidate record's value for every matching variable. Readers can regenerate and inspect it with `python scripts/21_provenance_sensitivity.py`.

B. A null that cannot be used

Permuting whole candidate rows does not provide a null for this statistic: maximum bipartite matching depends only on the multiset of candidate records, not on their order, so the statistic is invariant by construction. We implemented it, observed exactly zero variance, and replaced it with the copula null reported in the main text. The check is retained as `row_permutation_is_degenerate` in `provenance_sensitivity.py`, so the claim is verifiable rather than asserted. It is reported here, and not in the main

text, because it is a property of the estimator rather than a competing scientific null.

C. Which nulls are implemented, and whether they work

Two nulls are implemented and reported. *Independent column permutation* shuffles each candidate column separately, destroying every cross-variable relationship; beating it establishes only that candidate records are internally coherent. The *Gaussian-copula null* is the stricter one: within each outcome stratum the 13 containment variables are rank-transformed to normal scores, a correlation matrix is estimated on those scores, synthetic normal draws are generated through its Cholesky factor, and each column is mapped back through its own empirical marginal using nearest-value quantiles, so that every synthetic value is a value the column actually takes. Missingness is preserved per column and per record. Seeds derive from the master seed; 200 draws are used.

Nearest-value mapping matters. Linear interpolation would place synthetic values *between* the release’s discrete bin representatives, where no published interval can contain them; the null would then be depressed for a purely mechanical reason and would flatter the observed result. With the discrete support preserved, the null rises slightly, which is the conservative direction.

A null is only as good as its construction, so we measured whether it preserves what it claims (Table S3). Across the 78 variable pairs the mean absolute difference between the real and synthetic Spearman correlations is 0.031, against real correlations spanning -0.73 to $+0.87$; the largest single discrepancy is 0.168. No synthetic value falls outside its column’s observed support. The null therefore reproduces the cohort’s shape closely while destroying record identity, which is what makes exceeding it informative.

TABLE S3

NULL MODELS ACTUALLY IMPLEMENTED, AND WHETHER THE DEPENDENCE-PRESERVING NULL PRESERVES WHAT IT CLAIMS. ROW-ORDER PERMUTATION IS EXCLUDED: BIPARTITE MATCHING IS INVARIANT TO CANDIDATE ROW ORDER, SO IT IS A PROPERTY OF THE ESTIMATOR, NOT A NULL (SECTION S3-B).

Quantity	Value	Basis
Independent column permutation	0.003 ± 0.004	200 draws
exceedances $\geq$ observed	0	$p \leq 0.005$
Gaussian-copula null	0.059 ± 0.014	200 draws
exceedances $\geq$ observed	0	$p \leq 0.005$
Variable pairs compared	78	Spearman
Mean $ \text{real} - \text{null} $ correlation	0.031	retained
Largest single discrepancy	0.168	su bgr
Synthetic values off observed support	0	exact

S4. ALL REGISTRY-DERIVED CONFIGURATIONS

Table S4 reports every configuration derived from the corrected registry, with the full metric set. The main text shows seven of these; the remaining rows separate the two limbs of the KDIGO definition so that a reader can see each removed on

its own. *Saturated* marks a configuration whose lower interval bound is at or above 0.99. The two unsaturated rows are the ones restricted to information plausibly available before the renal work-up.

S5. CASE-MIX RE-WEIGHTING

Table S5 reports measured performance under target severity distributions, with the fitted models held fixed and the out-of-fold predictions re-weighted. It is a resampling exercise and not prospective evidence: it answers what would have been *measured* on a differently composed cohort using this model, not what a model trained on that cohort would achieve. The last column is the honest limit on it — with 9 stage-1 and 12 stage-2 cases in the benchmark, an early-heavy target reuses a few patients many times, and the Monte Carlo intervals do not reflect that.

S6. AUDIT FRAMEWORK: CANDIDATE FLAGGING VALUES

The main text reports the three measurements as quantities. Candidate flagging values are given here because they are calibrated on a single benchmark and should not be read as criteria:

- multivariable lift below 0.05,
- 90% of achievable gain reached below 25% of the training data,
- standardisation shift above 0.05.

Applied to this benchmark, at least one value is exceeded for every configuration. We repeat that this demonstrates internal consistency, not external validity: the values have not been shown to separate informative from uninformative benchmarks, and doing so requires a multi-dataset study with known ground truth that we have not performed.

A. The negative result, in full

Standardising ROC-AUC to a published community screening case mix moves it by at most 0.008 on this dataset — it would not flag a benchmark that Section V-B shows to be case-mix sensitive at a threshold. Standardising sensitivity at a fixed operating point moves it by up to 0.056. The explanation is specific to this sample: early-stage cases here have subgroup ROC-AUC 0.983 (21 cases), so re-weighting toward them changes few rankings. This is *not* evidence that AUC is generally insensitive to spectrum effects; a case mix that pushes cases below controls does move AUC. The practical implication is narrower: standardising discrimination alone can give false reassurance, so an operating-point metric should be standardised alongside it.

S7. EXTENSION TO SYNTHETIC TABULAR AND SYNTHETIC PATIENT DATA

The failure modes documented in the main text are not specific to real data, and the safeguards generalise to pipelines that generate or evaluate synthetic tabular records. Synthetic-data evaluation is harder in one respect: a generator trained on a source table can reproduce individual rows, so “is the

TABLE S4

EVERY REGISTRY-DERIVED CONFIGURATION, BEST MODEL PER ROW. 200 UNIQUE PATIENTS (128 CASES); 95% STRATIFIED PATIENT-LEVEL BOOTSTRAP INTERVALS. *Saturated* MARKS A CONFIGURATION WHOSE LOWER INTERVAL BOUND IS AT OR ABOVE 0.99, I.E. ONE RETAINING NO MEASURABLE HEADROOM ON THIS SAMPLE.

Feature set	<i>k</i>	Model	ROC-AUC (95% CI)	PR-AUC	Sens.	Spec.	Brier	Sat.
full_valid (reference)	25	SVM	1.000 (1.000–1.000)	1.000	1.000	1.000	0.004	yes
minus albuminuria information	24	SVM	1.000 (1.000–1.000)	1.000	1.000	1.000	0.004	yes
minus criterion inputs only (strict)	23	SVM	1.000 (1.000–1.000)	1.000	1.000	1.000	0.004	yes
minus all incorporation-risk variables	19	SVM	1.000 (1.000–1.000)	1.000	1.000	1.000	0.004	yes
minus creatinine/eGFR information	24	SVM	1.000 (1.000–1.000)	1.000	1.000	1.000	0.003	yes
minus consequences of advanced disease	21	SVM	0.998 (0.995–1.000)	0.999	0.969	0.986	0.017	yes
minus incorporation and consequences	15	SVM	0.994 (0.987–1.000)	0.997	0.969	0.972	0.023	no
conservative pre-index (no uncertain)	5	RF	0.930 (0.894–0.962)	0.968	0.852	0.875	0.095	no
pre-index available only	5	RF	0.930 (0.894–0.962)	0.968	0.852	0.875	0.095	no
minimal low-cost conservative	3	RF	0.881 (0.833–0.923)	0.944	0.773	0.764	0.124	no

TABLE S5

MEASURED PERFORMANCE UNDER TARGET SEVERITY DISTRIBUTIONS, MODELS HELD FIXED AND OUT-OF-FOLD PREDICTIONS RE-WEIGHTED.

EXPLORATORY RESAMPLING, NOT PROSPECTIVE EVIDENCE. INTERVALS ARE MONTE CARLO ONLY AND DO NOT REFLECT THE THINNESS OF THE SUPPORT: THE BENCHMARK HOLDS 9 STAGE-1 AND 12 STAGE-2 CASES, SO EARLY-HEAVY TARGETS REUSE A FEW PATIENTS MANY TIMES, AS THE LAST TWO COLUMNS SHOW.

Configuration	Target mix	Early frac.	ROC-AUC (MC 95%)	Sens.	Distinct / max reuse
low_cost	as sampled (benchmark)	0.16	0.993 (0.984–1.000)	0.968	81 / 8×
low_cost	community screening series	0.68	0.989 (0.981–0.996)	0.938	43 / 16×
low_cost	early-heavy	0.80	0.987 (0.979–0.994)	0.923	38 / 16×
low_cost	uniform across stages	0.40	0.990 (0.981–0.997)	0.950	75 / 10×
low_cost	advanced-heavy	0.00	0.994 (0.985–1.000)	0.979	74 / 8×
laboratory	as sampled (benchmark)	0.16	0.994 (0.986–0.999)	0.969	81 / 8×
laboratory	community screening series	0.68	0.985 (0.977–0.992)	0.912	43 / 16×
laboratory	early-heavy	0.80	0.984 (0.976–0.990)	0.879	38 / 16×
laboratory	uniform across stages	0.40	0.990 (0.983–0.996)	0.934	75 / 10×
laboratory	advanced-heavy	0.00	0.997 (0.990–1.000)	0.991	74 / 8×
full_valid	as sampled (benchmark)	0.16	1.000 (1.000–1.000)	1.000	81 / 8×
full_valid	community screening series	0.68	1.000 (1.000–1.000)	1.000	43 / 16×
full_valid	early-heavy	0.80	1.000 (1.000–1.000)	1.000	38 / 16×
full_valid	uniform across stages	0.40	1.000 (1.000–1.000)	1.000	75 / 10×
full_valid	advanced-heavy	0.00	1.000 (1.000–1.000)	1.000	74 / 8×

evaluation set independent of the training set?” becomes a question about memorisation as well as provenance.

Already implemented here, in a form adapted to real-data provenance: cryptographic dataset fingerprints (SHA-256 on every load, re-verified in continuous integration); exact-row and near-duplicate detection across training, validation, test and reference sets (the interval-containment matching, a tolerance-based near-duplicate test calibrated against a permutation null); entity-level overlap checks with held-out variable agreement; feature–target leakage scans and detection of target proxies and post-outcome variables (the registry and its guards); enforcement that all preprocessing is fitted inside cross-validation folds; provenance manifests recording origin, version, transformations and split history; and a fail-closed gate that blocks evaluation when a cohort is not verified independent.

Not implemented here, and required for synthetic data: membership-inference and memorisation checks, which ask

whether a synthetic record discloses that a particular real patient was in the training set; and nearest-neighbour similarity or distance-to-closest-record diagnostics, which ask whether synthetic rows are near-copies of real ones. The latter is both a privacy question and an evaluation-validity question — a synthetic test set whose records sit arbitrarily close to training records reproduces, in subtler form, exactly the non-independent evaluation this paper documents. The present work provides the provenance half of the problem, not the memorisation half.

A caution carries over with the recipe. The most consequential mechanism in this study was case composition, which no fingerprint, hash or duplicate scan would detect. A synthetic cohort can be free of memorisation, pass every overlap check, and still be uninformative because its severity distribution makes the task trivial. Provenance gating and informativeness diagnostics are therefore complementary rather than alternatives.

S8. REPRODUCIBILITY

A. One-command reproduction

`python reproduce.py`

validates input hashes, preprocesses, runs the analysis stages and test suite, regenerates tables and figures, builds the manuscript, and verifies that generated results match manuscript values. Individual targets are documented in `README.md`. A `Makefile` offers the same targets for POSIX users, but `make` is absent from the reference machine, so it is provided *unverified*: `reproduce.py` is the path we have run.

B. Environment

The reference environment is CPython 3.14.5 on Windows 11 with `numpy 2.5.2`, `pandas 3.0.5`, `scikit-learn 1.9.0` and `scipy 1.18.0`; the complete freeze is in `requirements.lock.txt`, which also records the BLAS build.

Identical seeds reproduce identical predictions within an environment. Across a rebuild from the same pinned versions, agreement is to about 10^{-16} rather than bit-for-bit, because floating-point summation order depends on the installed BLAS. The regression test asserts 10^{-12} rather than claiming exactness the project cannot deliver.

C. Data access

The two UCI releases are openly licensed (CC BY 4.0) and are retrieved by scripts with recorded checksums. The MIMIC-IV Clinical Database Demo is openly available from PhysioNet. One registered dataset — a community-based CKD screening series — is access-restricted and was not obtained; the manuscript states where that limits the work.

S9. REPORTING CHECKLIST

A completed TRIPOD+AI checklist ships as `reports/tables/table_36_tripod_ai.csv`: each item is marked satisfied, partly, adapted or not applicable, naming the section or artefact that addresses it. Items marked *partly* — eligibility, outcome timing, fairness, preregistration — are those the dataset cannot support.